GEOGRAPHY

INDIAN GEOGRAPHY





INDIAN CLIMATE

MODULE - 1

INDIAN MONSOON

- FACTORS DETERMINING THE CLIMATE OF INDIA

INDIAN MONSOON- AN INTRODUCTION

1. The climate of India is essentially hot-monsoonal climate, which is the prevalent climate in Asian countries.

2. The word monsoon has been derived from the Arabic word “mausim” which means season.

3. Originally, the word monsoon was used by Arab navigators, several centuries ago, to describe a system of seasonal reversal of winds along the shores of the Indian Ocean especially over the Arabian Sea.

4. Here, the winds blow from the southwest to northeast during summer season and from the northeast to southwest during the winter season.

5. In other words monsoons are periodic winds in which, there is a complete reversal of the wind direction.

6. South-east Asia, especially the sub-continent of India is the typical example of a monsoon region.

7. It is only around Indian Ocean that monsoonal circulation in true sense of the term is observed.

INDIAN MONSOON

UNITY AND THE DIVERSITY IN THE MONSOON CLIMATE

1. The monsoon regime emphasises the unity of India with the rest of South-east Asian region.
2. This view of broad unity of the monsoon type of climate should not, however, lead one to ignore its regional variations which differentiates the weather and climate of the different regions of India.
3. For example, the climate of Kerala and Tamil Nadu in the South are so different from that of Uttar Pradesh and Bihar in the North and yet all of these have a monsoonal type of climate.
4. The climate of India has many regional variations expressed in the pattern of winds, temperature and rainfall, rhythm of seasons and the degree of wetness or dryness.
5. These regional diversities may be described as subtypes of Indian monsoon climate.

INDIAN MONSOON

UNITY AND THE DIVERSITY IN THE MONSOON CLIMATE

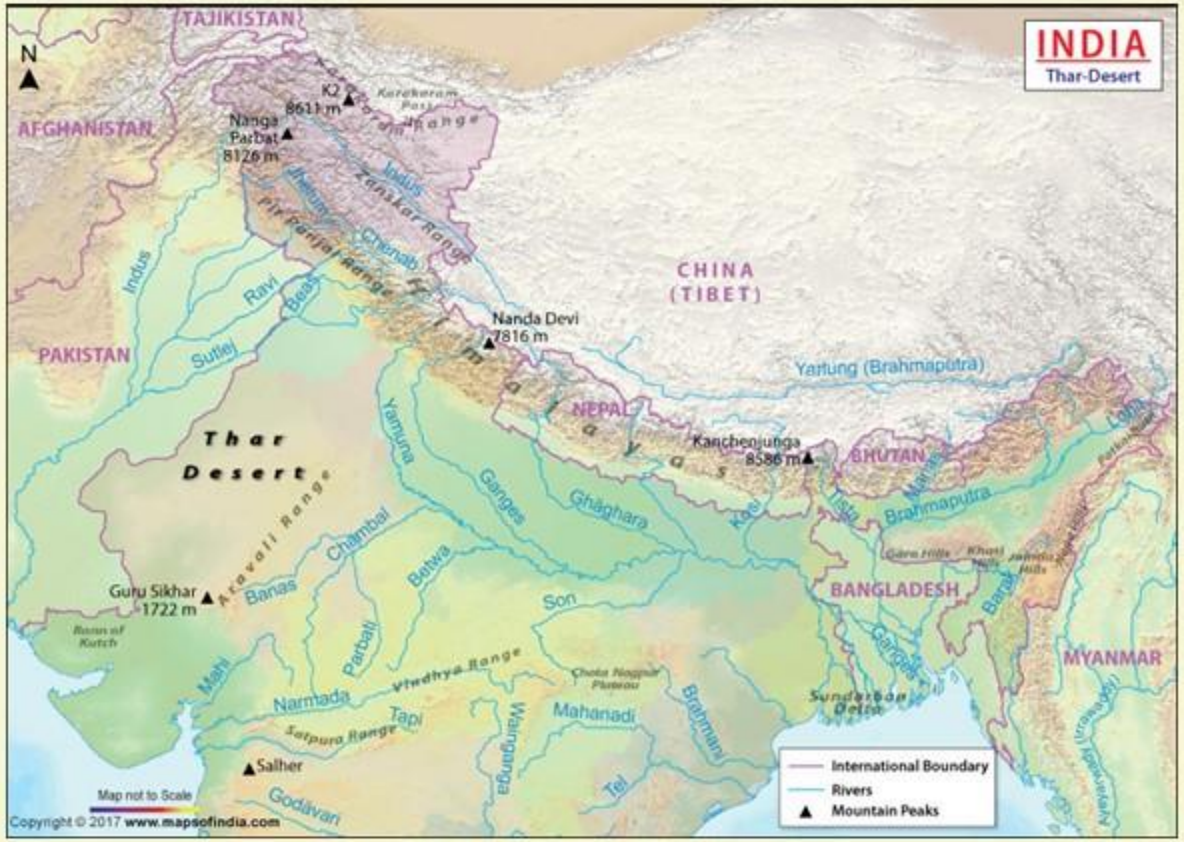
6 Let us take a closer look at these regional variations in temperature, winds and rainfall.

- Eg1: Churu in Rajasthan may record a temperature of 50°C or more on a June day, while the Mercury hardly touches 19°C in Tawang in Arunachal Pradesh on the same day.
- Eg2: On a December night, temperature in Dras in Jammu & Kashmir may drop down to -45°C, while Thiruvananthapuram or Chennai on the same night records 20°C or 22°C.


DRAS, JAMMU & KASHMIR
CHURU, RAJASTHAN
TAWANG, ARUNACHAL PRADESH
TRIVANDRUM, KERALA
IRAN
AFGHANISTAN
PAKISTAN
CHINA
NEPAL
BHUTAN
BANGLADESH
MYANMAR (BURMA)
OMAN
INDIA
Arabian Sea
Bay of Bengal

INDIAN MONSOON

UNITY AND THE DIVERSITY IN THE MONSOON CLIMATE



7 These examples conform that there are seasonal variations and temperatures change from place-to-place and from region-to-region in India.

8 Not only this, if we take only single place and record the temperature for just one day variations are no less striking.

9 Example, in the Thar desert if the day temperature is around 50°C at night it may drop down considerably up to 15°-20°C.

INDIAN MONSOON

UNITY AND THE DIVERSITY IN THE MONSOON CLIMATE

10 Now let us see the regional variations in precipitation. While snowfall occurs in the Himalayas, it only rains over the rest of the country.

11 Similarly, variations are noticeable not only in the type of precipitation but also in its amount.

12 E.g. 1: While Mawsynram in the Khasi hills of Meghalaya receive rainfall over 1,080 cm in a year, Jaisalmer in Rajasthan rarely gets more than 9 cm of rainfall during the same period.

INDIAN MONSOON

UNITY AND THE DIVERSITY IN THE MONSOON CLIMATE

13 E.g. 2: Tura situated in the Garo hills of Meghalaya may receive an amount of rainfall in a single day which is equal to 10 years of rainfall at Jaisalmer.

14 E.g. 3: The Ganga Delta and the coastal plains of Odisha are hit by strong rain bearing storms almost every 3rd or 5th day in July and August, while the Coromandel coast, a thousand kilometre to the south, goes generally dry during these months.

In spite of these differences and variations, the climate of India is monsoonal in rhythm and character.


IRAN
AFGHANISTAN
CHINA
PAKISTAN
JAISALMER, RAJASTHAN
NEPAL
MAWSYNRAM, MEGHALAYA
TURA, MEGHALAYA
OMAN
INDIA
MYANMAR
(BURMA)
Bay of Bengal
Arabian Sea

FACTORS DETERMINING THE CLIMATE OF INDIA

India's climate is controlled by a number of factors which can be broadly divided into two groups:

1. Factors related to location and relief
2. Factors related to air pressure and winds.

POINT NO:1

FACTORS RELATED TO LOCATION AND RELIEF

These factors related to location and relief can be further sub-divided into the following:

1. Latitude
2. The Himalayan mountains
3. Distribution of land and water
4. Distance from the sea
5. Altitude
6. Relief

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:1

LATITUDE

1. We know that the Tropic of Cancer passes through the Central part of India in east-west direction.

2. Thus, northern part of the India lies in sub-tropical zone and the part lying south of the tropic of cancer falls in the tropical zone.

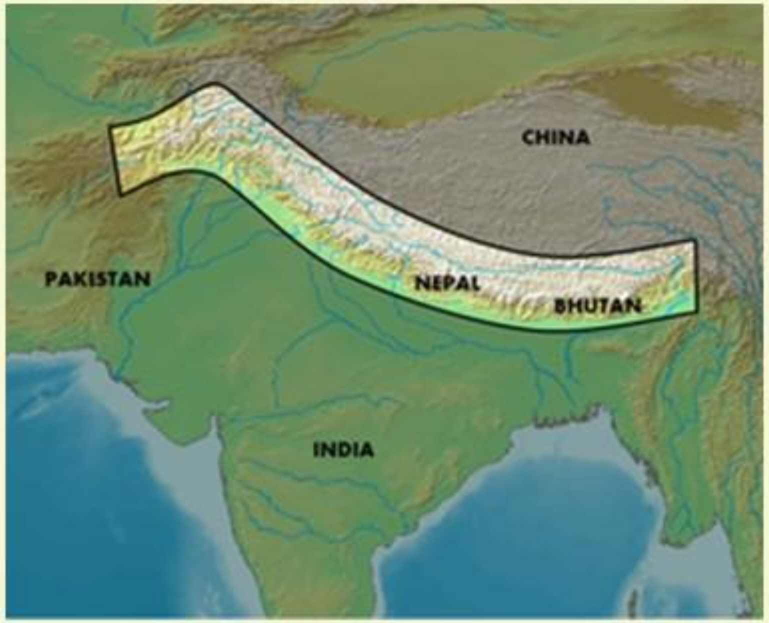
3. The tropical zone being nearer to the equator experiences high-temperatures throughout the year with small daily and annual range. Area north of the tropic of cancer being away from the equator experiences extreme climate with high daily and annual range of temperature.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:2

THE HIMALAYAN MOUNTAINS



1. The lofty Himalayas in the north along with its extensions act as an effective climatic divide.
2. The towering mountains provide an invincible shield to protect the sub-continent from the cold northern winds.
3. These cold and chilly winds originate near the Arctic circle and blow across central and eastern Asia.
4. The Himalayas also trap the monsoon winds forcing them to shed their moisture with in the sub-continent.

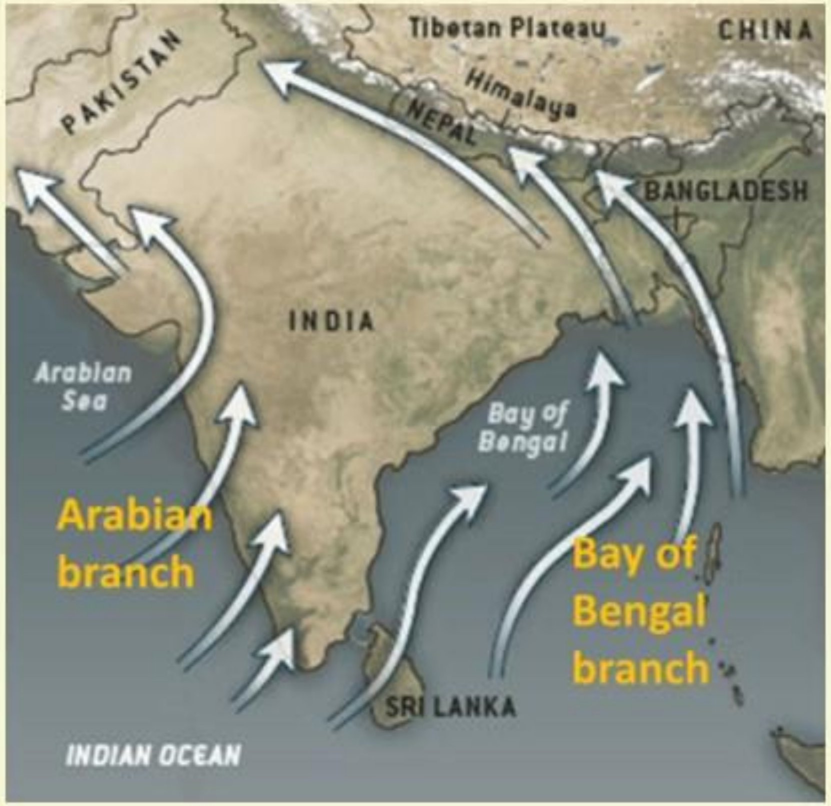
FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?



1) *Himalayas act as an effective physical barrier for moisture bearing southwest monsoon winds.*
2) *Himalayas divide the Bay of Bengal branch of monsoon winds into two branches_ one branch flowing along the plain region towards northwest India and the other towards southeast Asia.*
3) *If the Himalayas were not present the monsoon winds would simply move in to China and most parts of the Indian sub-continent would not have received sufficient amounts of rainfall as of now.*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?

4) *Himalayas protect India from the cold and the dry air masses, which originates from Siberia and Central Asian region, this would have entered the Indian sub-continent vowing to northeast trade winds thus making the place a cold dry desert with little or no vegetation.*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?

5) *In winters, the Himalayas bifurcate the sub-tropical westerly jetstreams winds and thus aid in the formation of winter-rainfall. Thus, Himalayas are responsible for fulfilling the water requirements of more than 1.3 billion population.*

6) *Without Himalayas, most of the place would have been a cold dry desert with little or no vegetation.*

7) *Also the racial profile of India would be very different as there would be very strong Mongol attacks because no natural defense would have been present and history might have been very different because of the extensive mongoloid invasion.*

UPSC PRELIMS QN

If there were no Himalayan ranges, what would have been the most likely geographical impact on India?

1. Much of the country would experience the cold waves from Siberia
2. Indo-Gangetic plain would be devoid of such extensive alluvial soils
3. The pattern of monsoon would be different from what it is at present

Which of the statements given above is/are correct?

a) 1 only
b) 2 & 3 only
c) 1 & 3
d) all of the above

For more questions related to this topic refer to Geography MCQ Module 4

EXPLANATION

- Answer : D
- India would be a cold desert without Himalayas
- Prone to Siberian Cold winds.
- Alluvial deposits brought down by Himalayan rivers would be lost.
- Diversion of South-West monsoon winds towards India will not take place and the Bay of Bengal branch will not get deflected towards us.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:3

DISTRIBUTION OF LAND AND WATER

1. India is flanked by the Indian Ocean on three sides in the south and it is having a high and a continuous mountain wall system in the north.
2. As compared to the land masses, water heats up or cools down slowly. This differential heating of land and sea creates different air pressure zones in different seasons in and around the Indian sub-continent.
3. Differences in air pressure causes reversal in the direction of monsoon winds.
4. Thus, the presence of land and water is also another important factor for the formation of Indian monsoon.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:4

DISTANCE FROM THE SEA

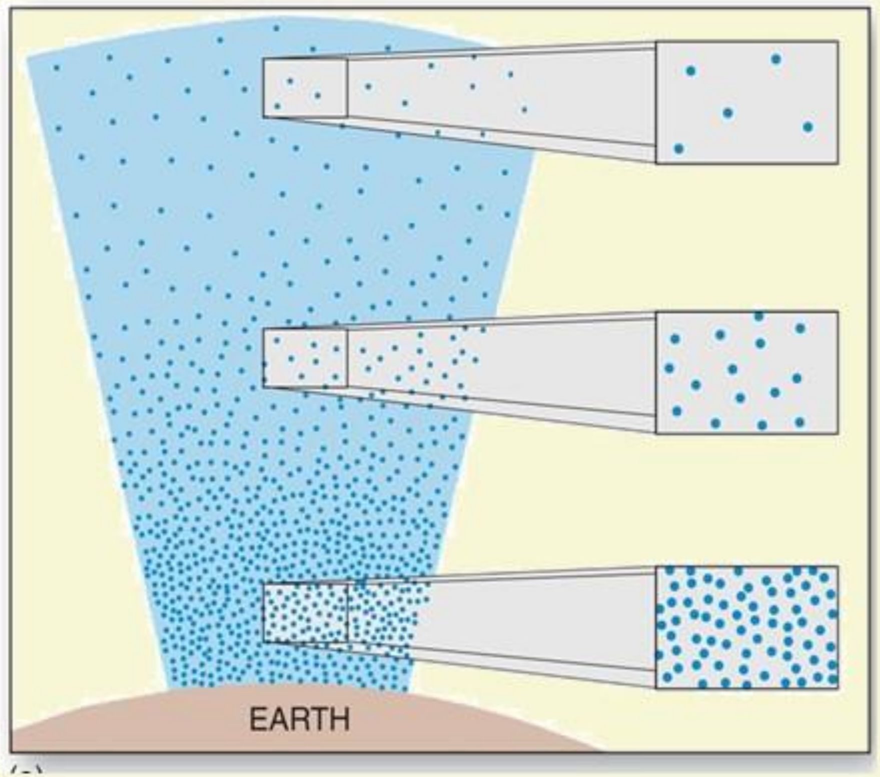
1. With a long coast line large coastal areas have an equable climate.
2. Areas in the interior of India are far away from the moderating influence of the sea, such areas have extremes of climate.
3. That is why, the peoples of Mumbai and the Konkan coast have hardly any idea of extremes of temperature and the seasonal rhythm of weather.
4. On the other hand the seasonal contrast in weather at places in the interior of the Country such as Delhi, Kanpur, and Amritsar affect entire the entire region and human life.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:5

ALTITUDE



1. Temperature decreases with height. Due to thin air places in the mountains are coolers than places on the plains.

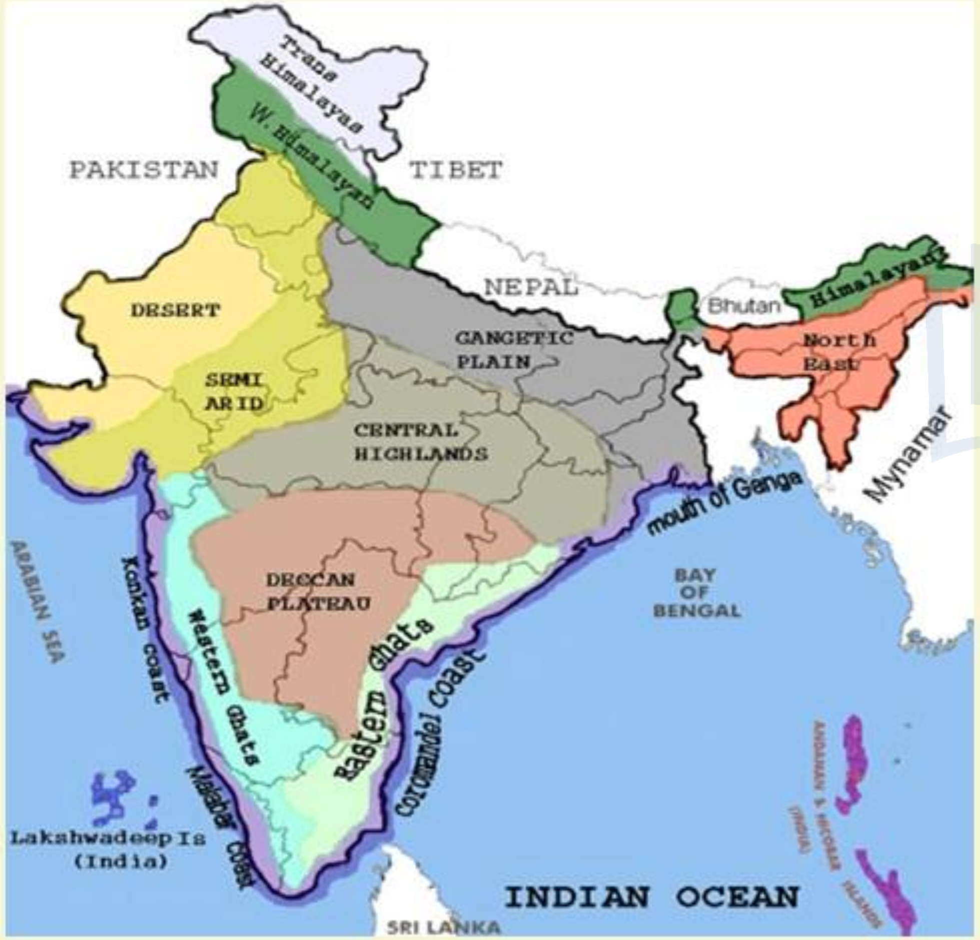
2. For example, Agra and Darjeeling are located on the same latitude but temperatures on January in Agra is 16°C whereas, it is only 4°C in Darjeeling.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

FACTOR NO:6

RELIEF



1. The physiography or relief of India also affects the temperature, air pressure, direction and speed of wind and the amount and distribution of rainfall.
2. The windward side of Western Ghats and Assam receive high rainfall during June to September, whereas, the southern plateau remains dry due to its leeward location along the Western Ghats.

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

Question- Why rainfall decreases from east to west in the Indo-Gangetic plain region?

1) *The Bay of Bengal branch of monsoon winds split into two branches - one branch flowing along the plain regions towards northwest India and other towards southeast Asia.*

2) *In summer there are many minor low-pressure cells that exist all over the plane region which eventually directs the monsoon winds from the eastern parts towards the western parts of India*

3) *As the monsoon winds move from east to west the moisture level decreases due to successive rainfall at each of the low pressure regions. By the time the wind reaches western parts of the plains all the moisture in the monsoon winds is exhausted and that is why rainfall decreases from east to west in plain region.*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

Question- Why is there no significant rainfall in Gujarat and Rajasthan?

1) *Monsoon winds flowing in Rajasthan and Gujarat are not obstructed by any orographic barrier, here the winds blow almost parallel to the Aravallis and hence these regions receive no rainfall.*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTORS RELATED TO LOCATION AND RELIEF

Question- Why are some parts in Peninsular India Semi-Arid?

1) *Places on the windward side of an orographic barrier receive great amount of rainfall, whereas, those on the leeward side remain arid to semi-arid due to rain shadow effect. This is how the southwest monsoon winds from the Arabian sea strike almost perpendicular at the Western Ghats and cause copious rainfall in the Western coastal plain and the western slopes of the Western Ghats.*

2) *On the contrary, vast areas of Maharashtra, Karnataka, Telangana, Andhra Pradesh, and Tamil Nadu lie in rain shadow or leeward side of the Western Ghats and receive scanty rainfall.*

3) *This is why some parts in the interiors of the Peninsular India are semi-arid.*

FACTORS DETERMINING THE CLIMATE OF INDIA

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